

Green Products

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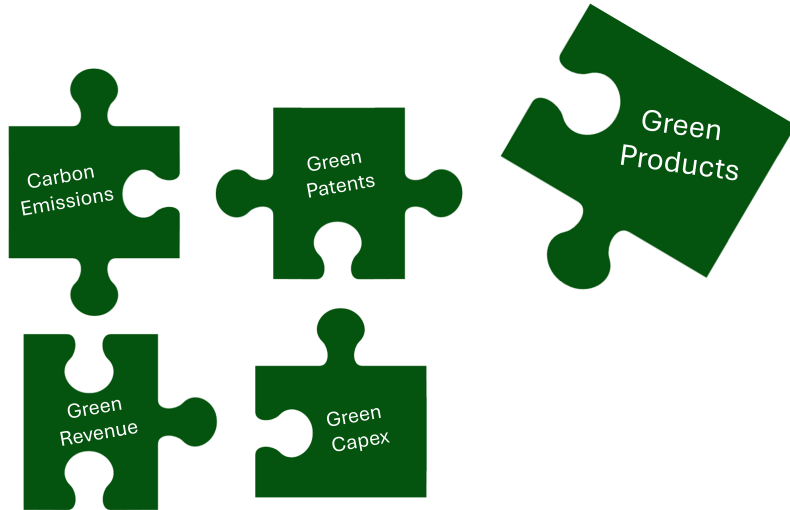
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Paper sheds light on an overlooked dimension in firms' net-zero transition



Three primary questions

1. Which firms are the active developers of green products?
 - ▶ Sectoral distribution: Manufacturing and energy sectors followed by service sector
 - ▶ Top 3 green groups: Energy production, energy conservation, & pollution control
2. Are firm's green products associated with its environmental and financial performance? If so, how?
 - ▶ Green products are associated with better environmental and financial performance
 - ▶ Particularly if there is synergy potential and little competition
3. Why do firms introduce green products?
 - ▶ Importance of externally created manager environmental awareness; e.g. through nearby natural disaster or peers' environmental scandals

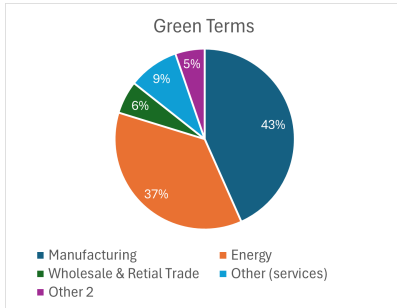
This discussion in two parts

Overarching suggestion: Dive into the unique features of the green trademarks dataset

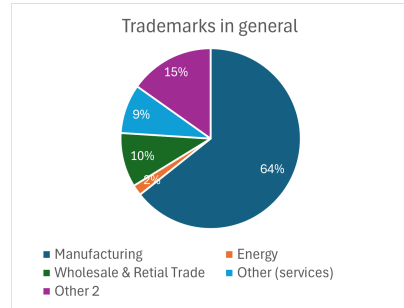
1. Additional context on new data point "green products"
2. Exploit unique dimensions of green-products:
 - a Environmental improvements associated with green products
 - b Complementarity of green and non-green terms

1. Additional context on new data point "green products"

Sectoral distribution of green product vs non-green product producers



Adapted from (Chiu, Hsu, LI & Tong, 2024); Manufacturing includes manufacturing and agriculture, Other (services) includes services and constructions, other 2 includes finance, transport and public utilities)



Adapted from (Desai, Gavrilova, Silva & Soares, 2024); Fama-Franch 12 industries grouped into Manufacturing [Manufacturing, Durables, Nondurables, chemicals, business equipment, healthcare], Energy [Oil, gas and coal], Other (services) [Other] and other 2 [Finance, telecommunications, utilities]

1. Additional context on new data point "green products"

Relationship of new products with sales growth and firm value

Table 5
Green products, sales growth, and firm value

	Sales Growth (next 2 years)		Total Q (next 2 years)		Tobin's Q (next 2 years)	
	(1)	(2)	(3)	(4)	(5)	(6)
Green Term	0.014*** (0.004)		0.041** (0.016)		0.206** (0.091)	
Green Category		0.011*** (0.002)		0.027** (0.012)		0.143** (0.067)
Non-Green Mark	0.012*** (0.002)	0.012*** (0.002)	0.040*** (0.010)	0.041*** (0.010)	0.172*** (0.057)	0.173*** (0.057)

- ▶ Green term = $\log(\# \text{ of green terms in newly registered green marks} + 1)$
- ▶ Green category = $\# \text{ of unique green categories in newly registered green marks} + 1$
- ▶ Non-green mark = $\log(\# \text{ of non-green marks} + 1)$

⇒ Financial impact seems to be very similar across green and non-green products

⇒ Is the role of product market scope, competitive threat, core vs non-core & greenfield products unique to green products?

2a. Environmental improvements associated with green products

Finding: Green products are associated with emission intensity improvements

Table 4
Green products and environmental performance

Panel B: Green products and GHG emissions

	GHG Total Emission (next 2 years)		GHG Scope1 Emission (next 2 years)	
	(1)	(2)	(3)	(4)
Green Term	-1.030** (0.508)		-0.777* (0.399)	
Green Category		-0.805** (0.332)		-0.606** (0.255)
Non-Green Mark	2.737 (1.998)	2.745 (1.999)	0.465 (0.297)	0.471 (0.296)

- ▶ Green term = $\log(\# \text{ of green terms in newly registered green marks} + 1)$
- ▶ Green category = $\# \text{ of unique green categories in newly registered green marks} + 1$
- ▶ Non-green mark = $\log(\# \text{ of non-green marks} + 1)$

⇒ Environmental impact is different for green and non-green terms

⇒ But how do green products and services lead to scope 1 emission improvements?

2a. Environmental improvements associated with green products

How do green products and services lead to scope 1 emission improvements?

- ▶ Negative relationship between green products and direct emissions is not obvious
 - ▶ Are green products also associated with process innovations?
 - ▶ Does green product innovation overlap with general efforts to reduce emissions?
 - ▶ Are green products substituted by non-green products and are these production processes less emission intensive?
 - ▶ Are green products a signal of a more general environmentally conscious management and approach?
 - ▶ Related, are green products part of a general emission reduction strategy driven by regulatory pressure?

2b. Complementarity of green and non-green terms

Finding: Combining green and non-green features increases the overall value of the product

Table 8
Dissecting green products: Green and non-green terms sharing common Nice class

	Sales Growth (next 2 years) (1)	Total Q (next 2 years) (2)	Tobin's Q (next 2 years) (3)
Common-Class Green Term	0.015*** (0.005)	0.047** (0.022)	0.214* (0.127)
Unique-Class Green Term	0.008* (0.005)	0.015 (0.018)	0.117 (0.103)
Firm FE	Y	Y	Y
SIC3-by-year FE	Y	Y	Y
Adjusted R-squared	0.305	0.591	0.641
Observations	90,401	90,401	90,401

- ▶ Common-Class = a non-green term with the same Nice class exists in the given green mark
- ▶ Unique-Class = non non-green term with the same Nice class exists in the given green mark

⇒ Within-product complementarity assessment

⇒ Hybrid vehicle product innovation more valuable than electric vehicle one?

2b. Complementarity of green and non-green terms

Why may combining green and non-green features be value-enhancing?

- ▶ Production synergies:
Do non-green features allow companies to exploit existing production know-how?
- ▶ Consumer appeal:
Do non-green features reach climate-conscious consumers without alienating those prioritizing non-environmental attributes?
- ▶ Incremental transition:
Do non-green features allows firms to lower risks associated with more radical green-only innovations?

2b. Complementarity of green and non-green terms

The broader role of green and non-green *product complementarity*

- ▶ Do green & non-green combinations serve as transition phase?
- ▶ What is the impact of green products for...
 - ▶ ... firms with mostly green products vs firms with mostly non-green products?
 - ▶ ... green vs non-green firms
 - ▶ ... firms in hard- vs easy-to-abate industries

Good luck with the paper

- ▶ Rich paper with fantastic data background work to identify green product and service innovations
- ▶ Many interesting relationships uncovered:
Can you dive deeper into the unique features of "green products"?
 - ▶ the complementarity of non-green terms and green-terms
 - ▶ the relationship between green products and future emissions